Chapter Evolution & Ecology Biology (Test) NUMS, PSC, MDCAT

1. The process by which changes in the genetic composition of populations of organisms occur in response to environmental changes is _____.

a) Inorganic evolution b) Biochemical Evolution c) Ecology d) Organic Evolution

2. The _____was the first person who argued from evidence that species were not specially created in their present forms.

a) Aristotle b) Lamarck c) Malthus d) Darwin

3. The Darwin proposed a mechanism for evolution, which he termed as _____

a) Artificial Selection b) Natural Selection c) Mutation d) Domestication

4. All living things came into existence in their present forms by Nature is _____

a) Natural Selection b) Big bang c) Creationism d) Evolution

5. Carolus Linnaeus was famous in the study of _____

a) divine creation b) Big bang c) Classification d) both a & c

6. The first concept of abiogenesis was given by _____

a) Aristotle b) Linnaeus c) Cuvier d) Darwin

7. Principle of Population was the publication of _____

a) Malthas b) Linnaeus c) Cuvier d) Darwin

8. The Cuvier theory of Catastrophism is supported by _____

a) Uniformitarianism b) Paleontology c) Sedimentation d) Both b & c

9. The _____published Principles of Geology.

a) Lyell b) Malthas c) Cuvier d) Darwin

10. The _____ wrote his essay on the origin of species.

a) Lyell b) Malthas c) Cuvier d) Darwin

11. The _____ of Linnaeus became a focal point in Darwin' s arguments for evolution.

a) Idea b) taxonomic system c) Phylogenetic system d) all

12. Archeobacteria support _____ theory.

a) Endosymbiotic b) Big Bang c) Catastrophism d) none

13. The first photosynthetic organisms produce _____ during photosynthesis.

a) O_2 b) Sugar c) S d) Both b & c

14. The first photosynthetic organisms used _____for reducing carbon dioxide.

a) H_2O b) $C_6H_{12}O_6$ c) H_2S d) S

15. The reducing atmosphere change to oxidizing atmosphere by _____

a) Algae b) Bacteria c) Plants d) Animals

16. About _____MYA protective ozone had built up to make life on land possible.

a) 120 b) 220 c) 320 d) 420

17. The prokaryotes may have arisen more than _____ years ago.

a) 3.5 billion b) 1.5 billion c) 3.5 million d) 1.5 million

18. The concept in which prokaryotes are changed into eukaryotes is _____

a) Endosymbiosis b) Big Bang c) Catastrophism d) Syncytial

19. The aerobic bacteria developed into mitochondria, concept coined by _____

a) Lyell b) Malthas c) Lynn Margulis d) None

20. The spirochetes is the longest bacteria which may change into _____

a) Algae b) Flagellum c) Cyanobacterium d) Dipnoi

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21. The mesosomes of bacterium resemble to _____
a) E.R b) Mitochondria c) Chloroplast d) Nucleus
22. The presence of _____ may support Lamarckism.
a) Appendix b) Ear muscles c) Eustachian Tubes d) all
23. On Galapagos island the Darwin study _____
a) Tortoise b) Finches c) Giraffe d) Both a & b
24. The _____was/were presented to the Linnaean Society of London on July 1, 1858.
a) Wallace Theory b) Darwin Theory c) Darwin-Wallace Theory d) none
25. The "Descent with Modification" by Darwin support _____
a) Monophyletic b) Polyphyletic c) Paraphyletic d) Holoblastic
26. Survival in the struggle for existence in life is_____
a) random b) not random c) not inherited d) by chance
27. The product of inherited variation is evolution and product of evolution is _____
a) Mutation b) Recombinant c) Species d) Generation
28. In 1940 the modern _____ theory was developed.
a) Mendel b) Darwin c) Wallac-Darwin d) Neo-Darwin
29. Neo-Darwin is a productive theory which integrate with _____
a) Paleontology b) Taxonomy c) Biogeography d) all
30. Darwin theory is supported by _____ evidences.
b) Paleontology b) Taxonomy c) Biogeography d) Both a & c
31. Armadillos, the armored mammals that live only in _____
a) Australia b) America c) Galapagos d) Africa island
32. The oldest known fossils are _____
a) Archaeopteryx b) Dipnoi c) Archeobacteria d) Pterodactyl
33. The sediment rocks have petrified parts by _____
a) Resin b) Silica c) $CaCO_3$ d) Both b & c
34. During evolution reptiles are changed into _____
a) Archaeopteryx b) Birds c) Mammals d) Both a & b
35. The organs differ in physiology but resemble in their anatomy are _____
a) Analogous b) Homologous c) Homozygous d) Isomorphic
36. Mammals with common anatomical theme like homology show _____
a) Monophyletic b) Polyphyletic c) Convergent evolution d) Divergent evolution
37. The oldest homologous structures are _____
a) Analogous b) vestibular organs c) homologous organs d) vestigial organs
38. Human recap the possible ancestor fish character evident by _____
a) To swim b) Gills c) Eustachian Tubes d) Single circuit Heart
39. The evolutionary most conservative molecule is _____
a) DNA b) RNA c) Cytochrome d) Bacteria
40. The evolution of antibiotic resistance in bacteria is _____
a) Artificial Selection b) Domestication c) Melanism d) Natural Selection
41. Total aggregate of genes in a population at any one time is called the population's _____
a) Genetic Drift b) Density c) gene pool d) pressure

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42. All alleles at all gene loci in all individuals of the population is _____
a) Genetic Drift b) Density c) gene pool d) pressure
43. If there are 500 plants then total of _____ genes for flower color in the population.
a) 500 b) 100 c) 1000 d) 50000
44. If the frequency of "A" is 0.6 then the frequency of "a" is _____
a) 0.3 b) 0.8 c) 0.4 d) 0.24
45. Shuffling of alleles due to meiosis and _____ has no effect on the overall genetic structure of a population.
a) random fertilization b) non-random fertilization c) Genetic Drift d) all
46. Individuals with P = 0.7 then their heterozygous offspring will be _____
a) 0.21 b) 0.42 c) 2.1 d) 4.2
47. If q = 0.3 then their homozygous recessive offspring will be _____
a) 0.21 b) 0.09 c) 0.42 d) 0.49
48. If P = 0.7 while q = 0.3 then their homozygous dominant offspring will be _____
a) 0.21 b) 0.09 c) 0.42 d) 0.49
49. The sum of all frequencies in a population if P = 0.7 while q = 0.3, _____
a) 0.21 b) 0.42 c) 0.49 d) 1
50. The _____ members of a population, cause disturbance in the gene pool.
a) Emigration b) Immigration c) Recombination d) Both a & b
51. When few individual fail to reproduce and genes are lost from population, it is _____.
a) Gene Flow b) Gene Pool c) Genetic Drift d) Emigration
52. The Natural Selection mainly depend upon _____
a) Acquired Characters b) Evolution c) Use/Disuse of organs d) Environment
53. In Pakistan. Cheetah especially white Cheetah is _____ species.
a) Threaten b) Endangered c) Extinct d) Fossil
54. In Pakistan, Habitat in peril create _____ species.
a) Threaten b) Endangered c) Extinct d) Fossil
55. A respiratory protein called_____ is found in all aerobic organisms.
a) Kinase b) Rubisco c) Cytochrome d) Enzyme
56. The term Ecology was coined by the German zoologist _____ in 1866.
a) Ernst Haeckel b) Robert Haeckel c) Charlous Haeckel d) E. Chaton
57. The _____ proposed a third kingdom Protista to accommodate Euglena like organisms and bacteria.
a) Ernst Haeckel b) Robert Haeckel c) Charlous Haeckel d) E. Chaton
58. The major unit of ecology is the _____
a) Abiotic Factors b) Biotic Factor c) ecosystem d) Environment
59. The various kinds of organisms that inhabit an ecosystem make up _____
a) Populations b) Environmental Biology c) Community d) Habitat
60. All populations within an ecosystem are known as a _____
a) Populations b) Environmental Biology c) Community d) Habitat
61. Major ecosystems, those that occupy broad geographical regions are called_____.
a) Universe b) biomes c) Ecosystem d) biosphere

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62. Biome consists of plants & animals in the fully developed _____ community.
a) Simple b) Complex c) Climax d) Pioneer
63. Combined the biome of earth together form the _____ ecosystem.
a) Planetary b) Biosphere c) Terrestrial d) Aquatic
64. The _____ is a thin layer of earth in which all living organisms exist.
a) Biome b) Biosphere c) Biota d) Biogeography
65. An organism lives in a _____
a) Home b) habitat c) Niche d) ecosystem
66. The actual location of place where an organism lives is called its _____
a) Home b) habitat c) Niche d) ecosystem
67. Ornithology is the study of _____
a) Fossils b) Birds c) Ornamental Plants d) Fish
68. Joseph Grinnell an American ornithologist first proposed the term _____
a) Home b) habitat c) Niche d) ecosystem
69. A species living under specific environmental conditions is _____
a) Home b) habitat c) Niche d) ecosystem
70. The _____is the ultimate distributional unit within which a species is restrained by the limitation of its physical structure and its physiology.
a) Home b) habitat c) Niche d) ecosystem
71. Charles Elton considered the _____, the basic role of an organism in the community what is does in and for living community.
a) Home b) habitat c) Niche d) ecosystem
72. Species's occupation within ecosystem is _____
a) Home b) habitat c) Niche d) ecosystem
73. The _____ includes all the physical factors of the environment necessary for survival.
a) Home b) habitat c) Niche d) ecosystem
74. The role a species plays in a community including behavior and influence is _____
a) Home b) habitat c) Niche d) ecosystem
75. When you are studying 50 to 100 plants of soybean in a field then it is _____
a) Autoecology b) Synecology c) Population d) Community
76. The study of the relationship of different populations is _____
a) Autoecology b) Synecology c) Population d) Community
77. While studying the community we study _____ levels of integration
a) individual b) population c) community d) all
78. Biosphere is spread out over the surface of the planet earth extending about _____km.
a) 8-10 b) 10-12 c) 12-14 d) 14-16
79. The earth is the components of _____
a) Atmosphere b) Hydrosphere c) Lithosphere d) Lithotripsy
80. Which one cover all_____
a) Atmosphere b) Hydrosphere c) Lithosphere d) Biosphere
81. Which one cover all_____
a) Atmosphere b) Ecosystem c) Biome d) Biosphere

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82. The bacteria are _____
a) Producer b) Decomposer c) Consumer d) both a & b
83. The major decomposer/reducer are _____
a) Fungi b) Bacteria c) Virus d) Decomposer
84. All the food chains and food webs begin with _____
a) Producer b) Decomposer c) Consumer d) Solar power
85. The trophic level 1 is _____
a) Producer b) Consumer c) Decomposer d) all
86. The herbivorous are at _____ trophic level.
a) 1 b) 2 c) 3 d) 4
87. The predators are at _____ trophic level.
a) 1 b) 2 c) 3 d) 4
88) The apex predators are at _____ trophic level.
a) 1 b) 2 c) 3 d) 4

89. The consumer are at _____ trophic level.

a) 2 b) 3 c) 4 d) all

90. The "community relay" in which plants and animals replace the earlier ones is _____

a) Success b) Biological succession c) Xerosere d) Derosere

91. In each case succession is initiated by a few hardy invaders called _____

a) Pioneers b) Community c) Climax d) Sere

91. Primary succession starting on dry rocks is _____

a) Hydrosere b) Xerosere c) Derosere d) lichen

93. The outer most rocky shell of soil is _____

a) Test b) Testa c) Crust d) Crustose

94. The pioneer of succession is _____

a) Crustose b) Folisose c) Fruiticose d) Climax

95. Permellia belongs to _____ lichen.

a) Crustose b) Folisose c) Fruiticose d) Climax

96. Which one cover all?

a) Polytrichum b) Tortula c) Bryopsida d) Bryophyta

97. The woody forest is considered to be the _____stage for this region.

a) Pioneers b) Community c) Climax d) Sere

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98. Which pair is mismatch in this set?

a) cat/mouse b) fox/rabbit c) seal/fish d) plasmodium/mosquito

99. The dandruff in hair is caused by _____

a) Dryness of skin b) Bacteria c) Fungi d) Non oily hair

100. All are not lichen except _____

a) Moss b) Spike moss c) Club moss d) Reindeer moss

101. Which contain bacteria?

a) Locule b) Lodicule c) Nodule d) Legume

102. The association between the roots of plants growing in acid soil and certain fungi is __

a) Endomycorrhizal b) Mycorrhizae c) EctoMycorrhizal d) all

103. Sparrow in our house make _____ relationship.

a) Predation b) Parasitism c) Commensalism d) Predation

104. Remoras make commensalism with _____

a) Ruminants b) Primate c) Cartilagenous fish d) Lion

105. The land used for grazing is _____

a) Crust b) Plain c) Pasture d) Forest

106. Over grazing may lead to the transformation of a grassland into a _____

a) Desert b) Pasture c) Desertification d) Aforestation

107. All are macronutrients except _____

a) H b) C c) Fe d) N

108. The chief reservoir of nitrogen is the _____

a) Atmosphere b) Plants c) Consumers d) Decomposers

109. Shortage of _____ in the soil is often the major limiting factor in plant growth.

a) H_2O b) CO_2 c) N d) Minerals

110. Several bacteria in soil are able to oxidize ammonia, it is called _____

a) Ammonification b) Nitrification c) Assimilation d) Denitrification

111. Several bacteria in soil are able to use protein and AAs then it is _____

a) Ammonification b) Nitrification c) Assimilation d) Denitrification

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112. The _____ is the form in which most nitrogen moves from the soil to roots.

a) Ammonia b) nitrate c) nitrites d) Fertilizer

113. The self – sustaining processes are _____

a) Photorespiration b) N-cycle d) Ammonia Solvay process d) all

114. The anaerobic bacteria break nitrates into nitrogen and oxygen is _____

a) Ammonification b) Nitrification c) Assimilation d) Denitrification

115. The total amount of energy fixed by plants is _____ productions.

a) gross primary b) primary c) net primary d) none

116. The amount of plant biomass is _____ production.

a) gross primary b) primary c) net primary d) none

117. The first organism during origin of life was _____.
a) chemoautotroph b) chemoheterotrophs c) Virus d) Autotroph

118. Homologous organ are those that show similarities in _____.
a) Appearance b) Function c) Origin d) Size

119. Which period is dubbed as the age of prokaryotes microbes _____.
a) Precambrian b) Palaeozoic c) Jurasic d) Mesozoic

120. Which one is not vestigial _____.
a) Coccyx b) Appendix c) Wisdom Teeth d) Nails

121. The process that has transformed life on earth from its earliest form is _____.
a) Mutation b) Evolution c) Migration d) Genetic drift

122. Important point/s of Lamarckism theory is/are
a) Use and disuse of organs b) Inheritance of acquired characters
c) Natural selection d) Both a & b

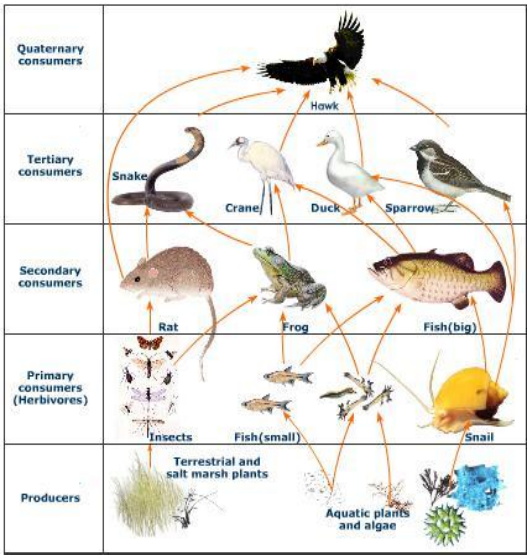
123. Neo-Darwinism has integrated ideas from _____.
a) Taxonomy b) Palaeontology c) Genetics d) All

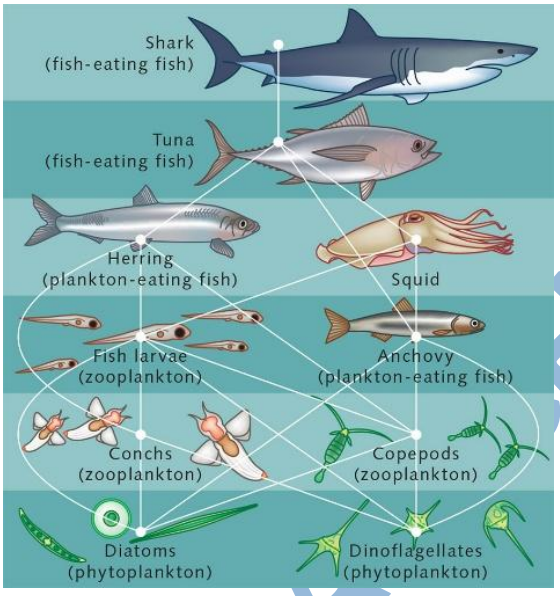
124. Most fossils are found in _____.
a) Soil. b) Sedimented rocks c) Resin d) Volcanic mountains

125. How many Trophic level are there in this figs?

a) 4 b) 5 c) 6 d) 10

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